

Scripts Index Table for ESEM 2025 Paper: "From Assessment to Enhancement of Pull Requests at Scale: Aligning Code Reviews with Developer Competencies Using Large Language Models"

Luca Mariotto

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1 Table

This section provides a central reference for the key scripts, data files, configuration files, and reports generated or utilized during this research. Generative AI tools provided assistance during the design and debugging phases for many of the scripts listed below. The complete collection of artifacts corresponding to this thesis has been archived on Zenodo and is available via the persistent DOI: <https://doi.org/10.5281/zenodo.15271232>. The table below indexes key items within this collection and provides links to browse specific files and directories in the associated public GitHub repository.

Table 1: Index of Research Artifacts

ID	Description	Type	File or Folder Name
S1	Scripts for fetching PR data from GitHub API	Script	<i>pr_fetcher_4.py</i>
S2	Scripts for automated PR scoring using zero-shot classification	Script	<i>pr_scorer.py</i>
S3	Scripts for identifying top-scoring PRs per competency	Script	<i>top_prs_getter.py</i>

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ID	Description	Type	File or Folder Name
S4	Script for PR quality analysis (readability, sentiment, counts)	Script	<i>pr_quality_analyzer.py</i>
S5	Script for comparing PR versions (similarity metrics)	Script	<i>pr_comparer.py</i>
S6	Script orchestrating model agnosticism analysis	Script	<i>analyze_agnosticism.py</i>
S7	Script for processing and analyzing survey results	Script	<i>survey_analysis_7.py</i>
S8	Script for holistic PR analysis (EDA, repo comparison, PCA, clustering)	Script	<i>pr_analyzer9.py</i>
S9	Survey load balancer script (JS/HTML)	Script	<i>index.html</i>
D1	Configuration for Latin Square design	Config	<i>data/config/SETUP.csv</i>
D2	Dataset of preprocessed and scored PRs	Data (Processed)	<i>data/raw/preprocessed_scored_prs.csv</i>
D3	Aggregated repository statistics (scores, counts)	Data (Processed)	<i>data/analysis_output/repository_mean_std_summary_filtered.csv</i>
D4	Repository correlation and clustering analysis data	Data (Processed)	<i>data/analysis_output/repository_summary_with_clusters_pca.csv</i>
D5	Combined processed survey responses	Data (Processed)	Not publicly available due to privacy commitments.
D6	Qualitative remarks extracted from survey	Data (Processed)	Not publicly available due to privacy commitments.
D7	Raw model agnosticism quality/similarity comparisons	Data (Raw)	<i>data/raw/enhanced_pr_comparison.csv</i>

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ID	Description	Type	File or Folder Name
D8	Summarized data analysis results	Data (Processed)	<i>data/analysis-output</i>
D9	Generated survey analysis reports and tables	Report/Data (Processed)	<i>data/reports</i>
D10	Model agnosticism statistical test report text	Report	<i>data/reports/report.txt</i>
D11	Raw PR quality analysis results	Data (Raw)	<i>data/raw/pr-quality-analysis.csv</i>
D12	Interactive clustering visualization	Data (Interactive)	<i>data/visualizations/ interactive_repo_clusters_sized_symbols.html</i>
P1	Prompt templates for generating PR variations	Prompt	<i>prompts</i>
V1	Additional Detailed Visualizations	Visualization	<i>data/visualizations</i>

Repositories Considered for Initial Fetching

The following list details the GitHub repositories initially identified and considered for fetching Pull Requests during the automated analysis phase. This broad selection aimed to cover diverse domains within software engineering. Note that filtering criteria (e.g., comment count, date range) and subsequent expert validation meant that the final 6 PRs used as stimuli in the human evaluation survey originated from a subset of these or potentially related repositories, and not all repositories listed here necessarily contributed usable data to the final analysis stages after filtering.

Table 2: List of GitHub Repositories Initially Considered for PR Fetching

Category	Owner	Repository
Automation & DevOps	ansible	ansible
Automation & DevOps	grafana	grafana
Automation & DevOps	jenkinsci	jenkins
Automation & DevOps	puppetlabs	puppet
Automation & DevOps	actions	runner

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Category	Owner	Repository
Automation & DevOps	saltstack	salt
Backend & Cloud Infrastructure	apache	cassandra
Backend & Cloud Infrastructure	containerd	containerd
Backend & Cloud Infrastructure	docker	cli
Backend & Cloud Infrastructure	elastic	elasticsearch
Backend & Cloud Infrastructure	apache	hadoop
Backend & Cloud Infrastructure	hashicorp	terraform
Backend & Cloud Infrastructure	istio	istio
Backend & Cloud Infrastructure	apache	kafka
Backend & Cloud Infrastructure	moby	moby
Backend & Cloud Infrastructure	nodejs	node
Backend & Cloud Infrastructure	pandas-dev	pandas
Backend & Cloud Infrastructure	prometheus	prometheus
Databases & Storage	cockroachdb	cockroach
Databases & Storage	etcd-io	etcd
Databases & Storage	influxdata	influxdb
Databases & Storage	mariadb	server
Databases & Storage	mongodb	mongo
Databases & Storage	postgres	postgres
Databases & Storage	redis	redis
Databases & Storage	facebook	rocksdb
Development Tools	bazelbuild	bazel
Development Tools	eslint	eslint
Development Tools	git	git
Development Tools	neovim	neovim
Development Tools	openresty	openresty
Development Tools	prettier	prettier
Game Engines & Graphics	blender	blender
Game Engines & Graphics	godotengine	godot
Game Engines & Graphics	openframeworks	openFrameworks
Game Engines & Graphics	pygame	pygame
Game Engines & Graphics	defold	defold
Machine Learning & AI	facebookresearch	detectron2
Machine Learning & AI	fastai	fastai
Machine Learning & AI	scikit-learn	scikit-learn
Machine Learning & AI	tensorflow	tensorflow
Machine Learning & AI	huggingface	transformers
Machine Learning & AI	pytorch	pytorch
Networking	cilium	cilium

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Category	Owner	Repository
Networking	coredns	coredns
Networking	zephyrproject-rtos	zephyr
Operating Systems & Low-Level Software	freebsd	freebsd
Operating Systems & Low-Level Software	openbsd	src
Operating Systems & Low-Level Software	qemu	qemu
Operating Systems & Low-Level Software	reactos	reactos
Operating Systems & Low-Level Software	systemd	systemd
Operating Systems & Low-Level Software	xen-project	xen
Programming Languages	openjdk	jdk
Programming Languages	python	cpython
Programming Languages	golang	go
Programming Languages	llvm	llvm-project
Programming Languages	rust-lang	rust
Programming Languages	apple	swift
Programming Languages	microsoft	TypeScript
Security & Cryptography	letsencrypt	boulder
Security & Cryptography	sigstore	cosign
Security & Cryptography	libressl	libressl
Security & Cryptography	openssl	openssl
Security & Cryptography	hashicorp	vault
Web & API Frameworks	aspnet	aspnetcore
Web & API Frameworks	django	django
Web & API Frameworks	expressjs	express
Web & API Frameworks	fastify	fastify
Web & API Frameworks	pallets	flask
Web & API Frameworks	rails	rails
Web & API Frameworks	spring-projects	spring-framework
Web & Frontend Frameworks	angular	angular
Web & Frontend Frameworks	emberjs	ember.js
Web & Frontend Frameworks	flutter	flutter
Web & Frontend Frameworks	vercel	next.js
Web & Frontend Frameworks	facebook	react
Web & Frontend Frameworks	solidjs	solid
Web & Frontend Frameworks	sveltejs	svelte
Web & Frontend Frameworks	tailwindlabs	tailwindcss
Web & Frontend Frameworks	vuejs	vue